

Question 1 - Find Maximum and Minimum

START

Read array

max = first element

min = first element

FOR each element from index 1 to end

IF element > max

max = element

IF element < min

min = element

PRINT max

PRINT min

END

Question 2 — Second Largest Element

START

Read array

largest = first element

secondLargest = no value yet

FOR each element from index 1:

IF element > largest:

 secondLargest = largest

 largest = element

ELSE IF element < largest AND

 (secondLargest does not exist OR element > secondLargest):

 secondLargest = element

IF secondLargest does not exist:

 print "No second largest element"

ELSE:

 print secondLargest

END

Question 3 — Move All Zeros to the End

START

index = 0

FOR each element in array:

 IF element is not zero:

 arr[index] = element

 index++

WHILE index < array length:

arr[index] = 0

index++

END

Question 4: Student Queue Management

START

Create empty ArrayList called queue

REPEAT

Display menu

Read choice

IF choice = 1

Read student ID

Add student ID to end of queue

ELSE IF choice = 2

IF queue is empty

Display "Queue is empty"

ELSE

Remove student at index 0

Display student ID

ELSE IF choice = 3

Read student ID

Search student in queue

Display whether found or not

ELSE IF choice = 4

Display queue

ELSE IF choice = 5

Display queue size

ELSE IF choice = 6

Exit

ELSE

Display invalid choice

UNTIL choice = 6

END